STOC: Optimal Experimental Design

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Introduction

OED

Optimal Experimental Design (or OED for short) is a method for designing experiments that aims at maximising information gain while minimising a cost function. This cost function can encompass uncertainty, cost, time… it is up to the user to select the parameters to optimise.

The key concept is that **not all observations are equally informative**.

This concept[1] can be particularly interesting when working with participatory science datasets, which is the case of the **STOC** dataset, which we are using here.

BIODICAPT

BIODICAPT is a French initiative that aims at monitoring biodiversity of agricultural lands on a large (*aka national*) scale through the use of various recording devices.

BIODICAPT started in early-2026, the first results (data extraction of species distributions) will not be available until late-2026 or early-2027. That is why, in the meantime, we are going to use **STOC** to prepare our workflow.

Strategy

With OED, the question we want to answer is simple: **which points are most informative in the 500ENI network, given our previous data collection and our constraints?**

For that, we'll use explicative variables collected on site (temperatures, precipitation, type of land, …), response variables obtained from recorders (presence-absence of birds species mainly), and Species Distribution Models (SDM), in particular HMSC, which is a joint-SDM (JSDM) [1]. These models make use of several species distributions to find *latent* links between them, which supposedly improves the prediction accuracy.

They are opposed to single SDMs (Species Distribution Models applied to one species) and SSDMs (Stacked SDMs, are mutliple SMDs glued together, with no link between them).

HMSC

HMSC (Hierarchical Modelling of Species Communities) is a JSDM developped by Ovaskainen et al. (2017) [1], it is distributed as an open-source R-package on GitHub [2].

It is undoubtebly the most widely used JSDMin the last years. To summarise its principle, take a quick look at Figure 4, from their original publication [1]:

[1] Sometimes, also referred to as "adaptive sampling".

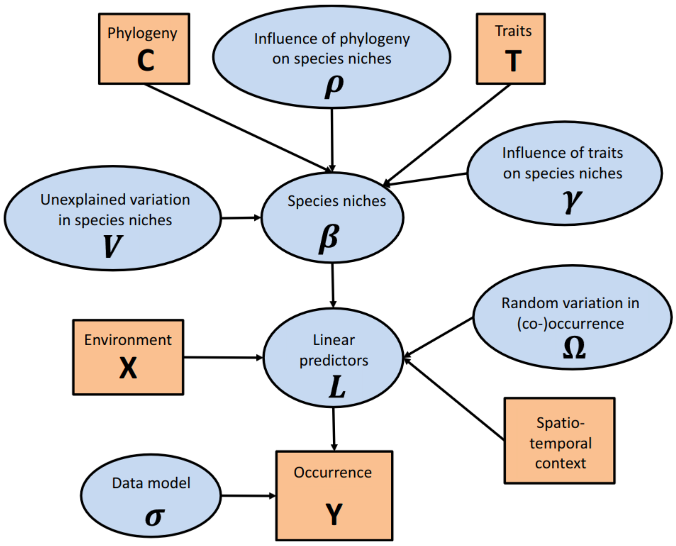


Figure 4 A graphical summary of the HMSC statistical framework. In this Directed Acyclic Graph (DAG), the orange boxes refer to data, the blue ellipses to parameters to be estimated, and the arrows to functional relationships described with the help of statistical distributions.

Figure 1: Copy of Figure 4 from Ovaskainen et al. (2017) [1]

GJAM

Work in progress...

Only HMSC has been implemented into our code for now. GJAM is under consideration, as well as other SDMs: RF, SSDM, deep-SDMs, ...
This section will be updated when new results are available.

Materials & Methods

We use a dataset that is already curated and known to work with HMSC: **STOC** [3].

The French Breeding Bird Survey (FBBS, or STOC in french) is a standardised multi-species abundance dataset, based on a citizen science program. It contains bird sightings and explanatory/environmental variables. Since it is already pre-processed, we can work directly with it, out of the box.

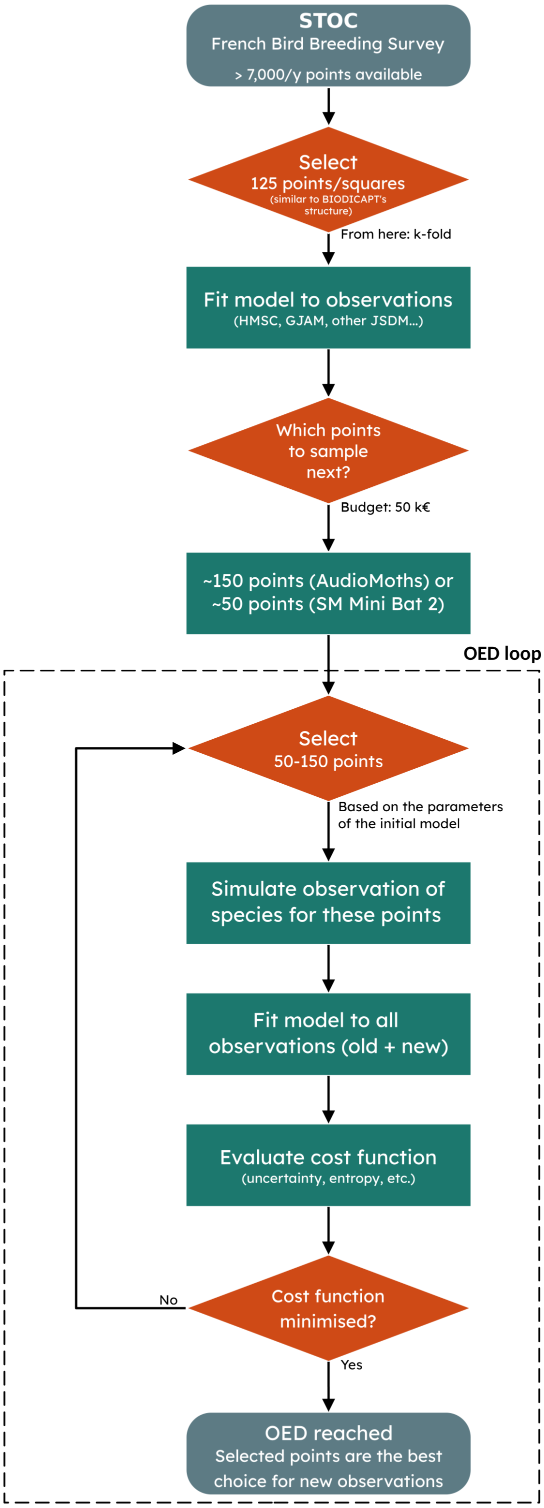


Figure 2: Flowchart of the workflow with STOC dataset

Results

Adding new points

First, we want to make sure that, following the flowchart above, we can actually make better models by adding (50-150) new points to our dataset.

Reference results

To test this, we draw on *Mondain-Monval et al.* work [4]. However, it is only a base on which we build, not an exact replica as our method and dataset are quite different from the original *Mondain-Monval et al.* publication:

Parameter	M.M.	Ours
Dataset	Simulated with `virtualspecies` package	STOC annual sampling
Number of species	50 species	15 species
Number of variables	10 (randomly selected from 33 available)	6 (direct from STOC)
Model type	Ensemble model (GLM + GAM + RF)	HMSC
Re-training strategy	Re-sampling of 1%, 10% or 50% of dataset	Addition of new samples (50)
Training size	2000 cells	125 cells + new data

With this context in mind, here are their results:

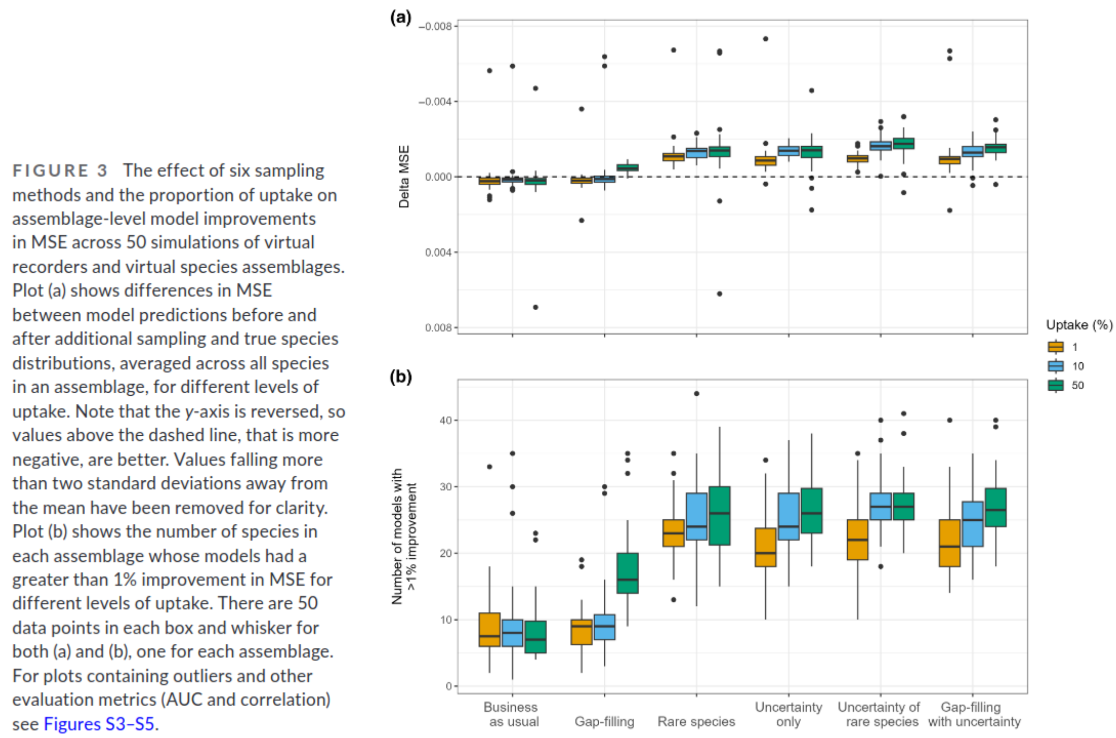


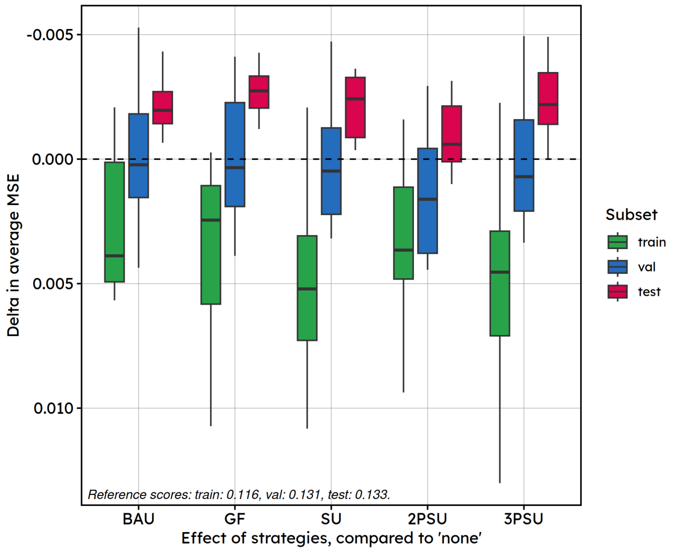
Figure 3: Copy of Figure 3a mondain-monval et al., 2024

Our results

How much does the model improve when adding 50 new samples to the *base* dataset (125 samples)?

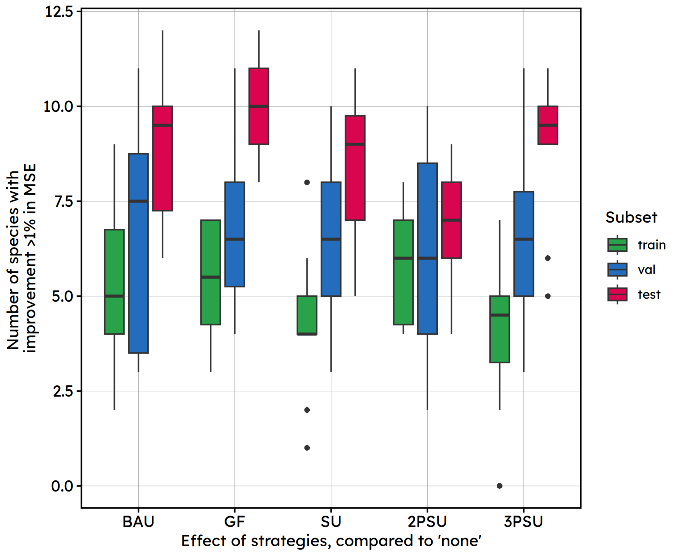
We investigate this depending on different strategies.

Boxplot differences



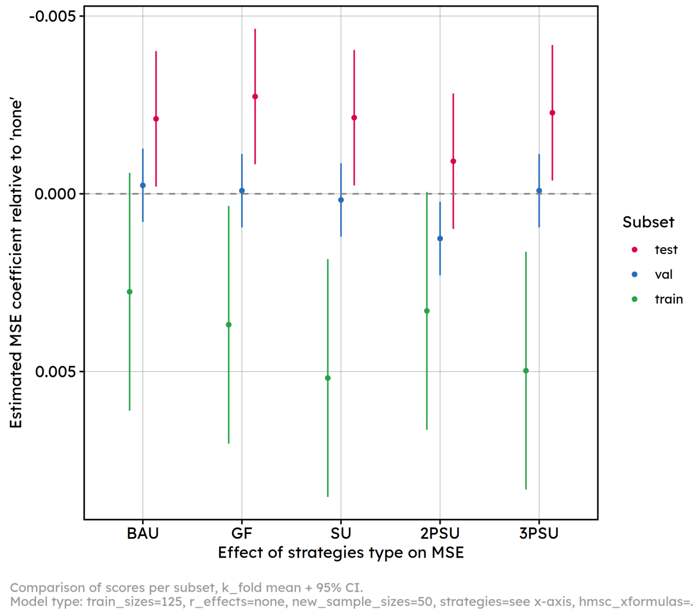
Distribution of per-fold differences (species-averaged within each fold, across k_fold).
Reference: train_sizes=125, r_effects=none, new_sample_sizes=50, strategies=none, hmsc_xformulas=6..
Compared with: train_sizes=125, r_effects=none, new_sample_sizes=50, strategies=see x-axis, hmsc_xformulas=6.

Boxplot improvement



Number of species per box: 15 .
Reference: train_sizes=125, r_effects=none, new_sample_sizes=50, strategies=none, hmsc_xformulas=6..
Compared with: train_sizes=125, r_effects=none, new_sample_sizes=50, strategies=see x-axis, hmsc_xformulas=6.

Estimated effects



BAU: Business-as-usual; GF: Gap-filling; SU: Simplified-uncertainty; XPSU: SU divided in X parts

Fitted model:

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: score ~ loop_element + (1 | k_fold) + (1 | species)
   Data: subset(scores_df, dataset == subset)

REML criterion at convergence: -5837.3

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.0297 -0.6775 -0.0617  0.6780  3.9248

Random effects:
 Groups   Name        Variance  Std.Dev.
 species  (Intercept) 5.831e-03 0.076364
 k_fold   (Intercept) 6.421e-06 0.002534
 Residual             7.074e-05 0.008411
Number of obs: 900, groups:  species, 15; k_fold, 10

Fixed effects:
                                            Estimate Std. Error         df
(Intercept)                                1.331e-01  1.975e-02  1.407e+01
loop_elementbusiness-as-usual             -2.109e-03  9.712e-04  8.710e+02
loop_elementgap-filling                   -2.737e-03  9.712e-04  8.710e+02
loop_elementsimplified-uncertainty        -2.143e-03  9.712e-04  8.710e+02
loop_element2-part-simplified-uncertainty -9.166e-04  9.712e-04  8.710e+02
loop_element3-part-simplified-uncertainty -2.280e-03  9.712e-04  8.710e+02
                                          t value Pr(>|t|)
(Intercept)                                 6.742 9.18e-06 ***
loop_elementbusiness-as-usual              -2.172  0.03013 *
loop_elementgap-filling                    -2.818  0.00494 **
loop_elementsimplified-uncertainty         -2.206  0.02763 *
loop_element2-part-simplified-uncertainty  -0.944  0.34553
loop_element3-part-simplified-uncertainty  -2.348  0.01911 *
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusions

These results show that, when adding 50 new points to our training set (125 points), training, test and validation scores vary depending on the strategy chosen to add those points.

In details "BAU", "GF" and "SU" improve significantly the results in test (compared to the "base" model which is not adding new points), while worsening results in training (expected since unknown points are added to the training set).

Interestingly, adding new samples on several iterations (i.e. in 2 times with "2PSU", or 3 times with "3PSU") instead of all at once (i.e. "SU") yields to differences in scores. It seems like "2PSU" scores are closer to the one of the base strategy, while "3PSU" scores show improvement comparable to "SU" and "GF" (clearer on this plot).

Overall however, these effects sizes (~0.0025 in MSE units) are small relative to the residual noise (SD 0.0086) and an order of magnitude smaller than between-species variability (SD 0.076).

Comparison with Mondain-Monval et al.

Mondain-Monval et al. [4] and we computed the difference in MSE between a base strategy (business-as-usual) and other strategies. However, our training sets were not built in the same way: - Mondain-Monval et al. [4] **changed a proportion of their training set** (1%, 10%; 50%) according to the chosen strategy. - We **added new samples to our training set** (50, with a base strategy of 125 samples) according to the chosen strategy.

Our figures also show scores in train/val/test. We assume Mondain-Monval et al. [4] show only scores in test.

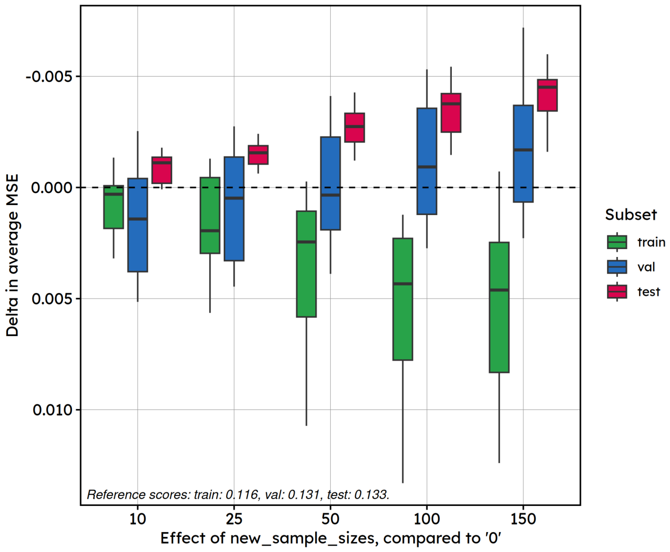
In our results, differences in MSE between resampling strategies are slight, less marked than those of [4]. Statistical tests reveal that there are significant differences in the test scores compared to no resampling. Interestingly, the "SU" strategy does not clearly yields to the best results (which was what we expected from [4]).

An interesting continuation is to look at the effect of adding more or less points, depending on the strategy. We chose "GF", "SU" and "3PSU" to investigate this further.

How many points to add?

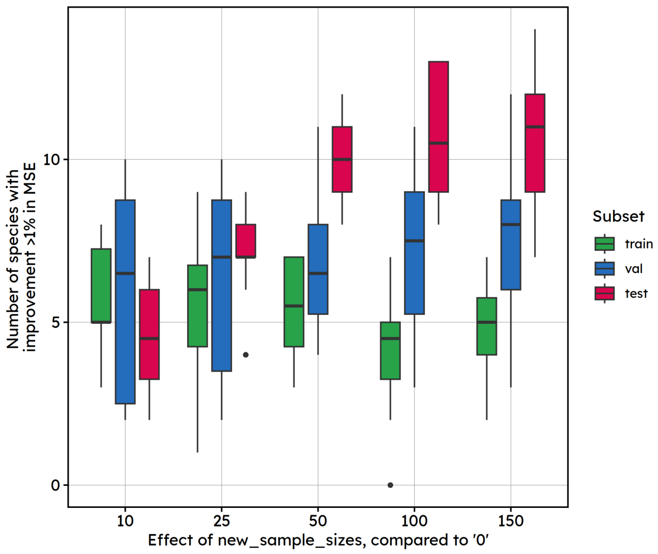
Effect using "GF"

Boxplot differences



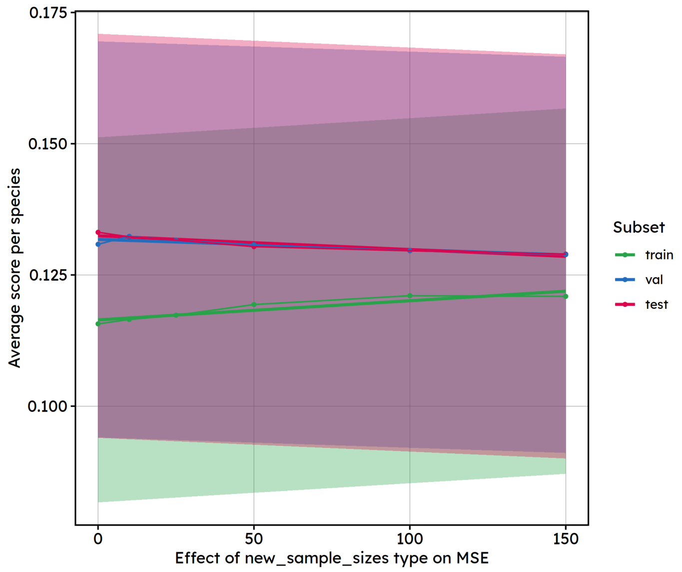
Distribution of per-fold differences (species-averaged within each fold, across k_fold).
Reference: train_sizes=125, r_effects=none, new_sample_sizes=0, strategies=gap-filling, hmsc_xformulas=6..
Compared with: train_sizes=125, r_effects=none, new_sample_sizes=see x-axis, strategies=gap-filling, hmsc_xformulas=6.

Boxplot improvement



Number of species per box: 15 .
Reference: train_sizes=125, r_effects=none, new_sample_sizes=0, strategies=gap-filling, hmsc_xformulas=6..
Compared with: train_sizes=125, r_effects=none, new_sample_sizes=see x-axis, strategies=gap-filling, hmsc_xformulas=6.

Estimated effects



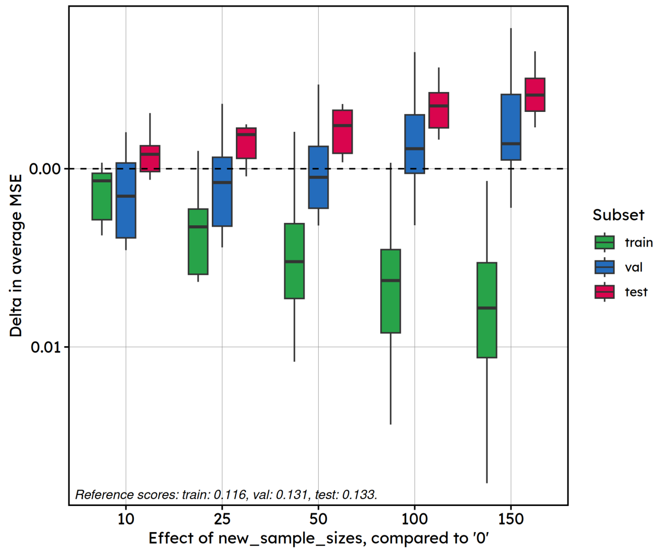
Comparison of scores per subset, k_fold mean + 95% CI.
Model type: train_sizes=125, r_effects=none, new_sample_sizes=see x-axis, strategies=gap-filling, hmsc_xformulas=.

Fitted model:

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: score ~ loop_element + (1 | k_fold) + (1 | species)
   Data: subset(scores_df, dataset == subset)

REML criterion at convergence: -5899.3

Scaled residuals:
    Min      1Q  Median      3Q     Max
-2.7947 -0.6888 -0.0971  0.6497  3.2451

Random effects:
 Groups   Name        Variance  Std.Dev.
 species  (Intercept) 5.757e-03 0.075874
 k_fold   (Intercept) 6.594e-06 0.002568
 Residual             6.862e-05 0.008284
Number of obs: 900, groups:  species, 15; k_fold, 10

Fixed effects:
               Estimate Std. Error         df t value Pr(>|t|)
(Intercept)   1.325e-01  1.961e-02  1.405e+01   6.754 9.08e-06 ***
loop_element -2.614e-05  5.184e-06  8.750e+02  -5.042 5.59e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

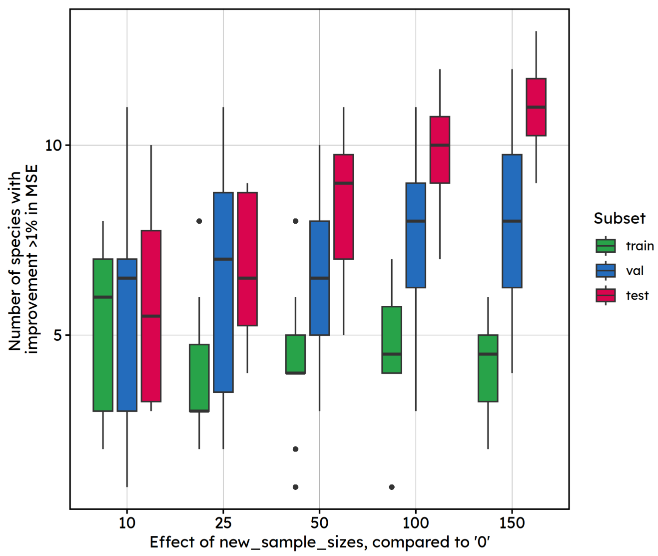
Effect using "SU"

Boxplot differences



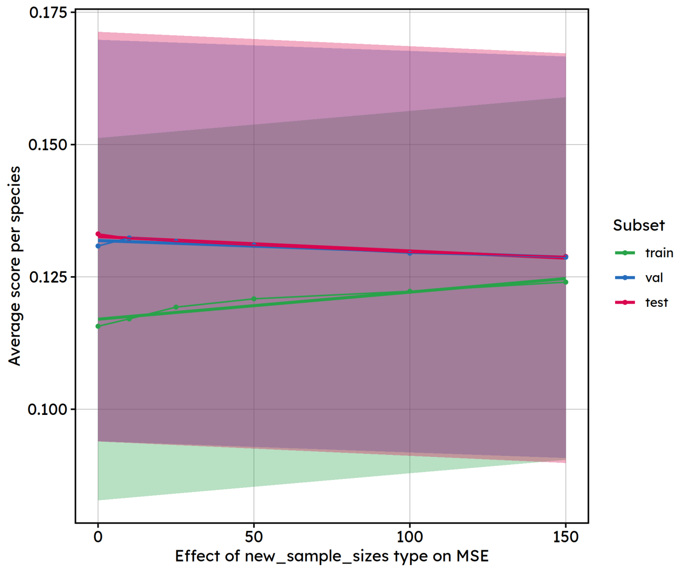
Distribution of per-fold differences (species-averaged within each fold, across k_fold).
Reference: train_sizes=125, r_effects=none, new_sample_sizes=0, strategies=simplified-uncertainty, hmsc_xformulas=6..
Compared with: train_sizes=125, r_effects=none, new_sample_sizes=see x-axis, strategies=simplified-uncertainty, hmsc_xformulas=6.

Boxplot improvement



Number of species per box: 15 .
Reference: train_sizes=125, r_effects=none, new_sample_sizes=0, strategies=simplified-uncertainty, hmsc_xformulas=6..
Compared with: train_sizes=125, r_effects=none, new_sample_sizes=see x-axis, strategies=simplified-uncertainty, hmsc_xformulas=6.

Estimated effects



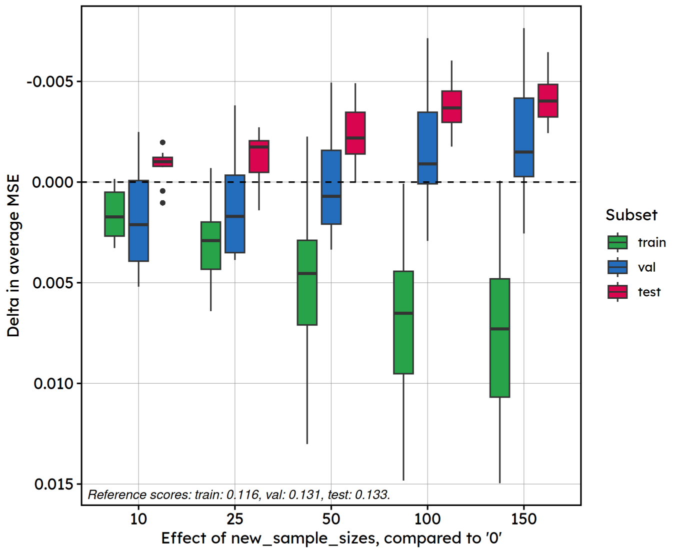
Comparison of scores per subset, k_fold mean + 95% CI.
Model type: train_sizes=125, r_effects=none, new_sample_sizes=see x-axis, strategies=simplified-uncertainty, hmsc_xformulas=.

Fitted model:

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: score ~ loop_element + (1 | k_fold) + (1 | species)
   Data: subset(scores_df, dataset == subset)

REML criterion at convergence: -5853

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.1231 -0.6802 -0.0923  0.6420  3.3094

Random effects:
 Groups   Name        Variance  Std.Dev.
 species  (Intercept) 5.819e-03 0.076283
 k_fold   (Intercept) 5.916e-06 0.002432
 Residual             7.241e-05 0.008509
Number of obs: 900, groups:  species, 15; k_fold, 10

Fixed effects:
               Estimate Std. Error         df t value Pr(>|t|)
(Intercept)   1.326e-01  1.972e-02  1.405e+01   6.726 9.51e-06 ***
loop_element -2.713e-05  5.326e-06  8.750e+02  -5.095 4.27e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

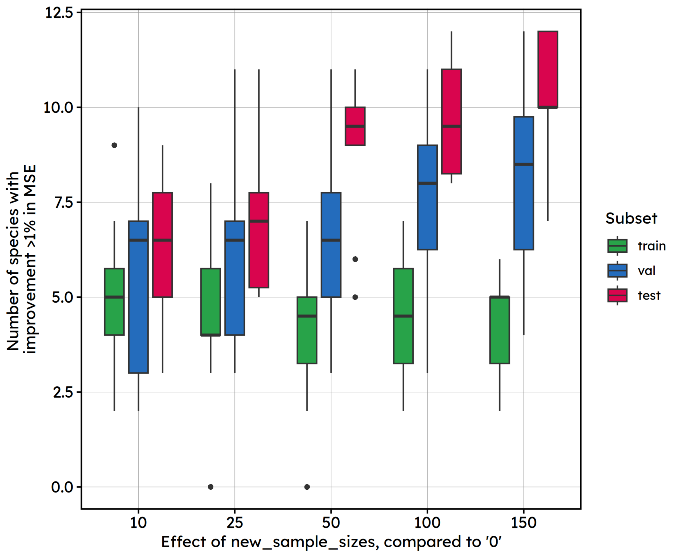
Effect using “3PSU”

Boxplot differences



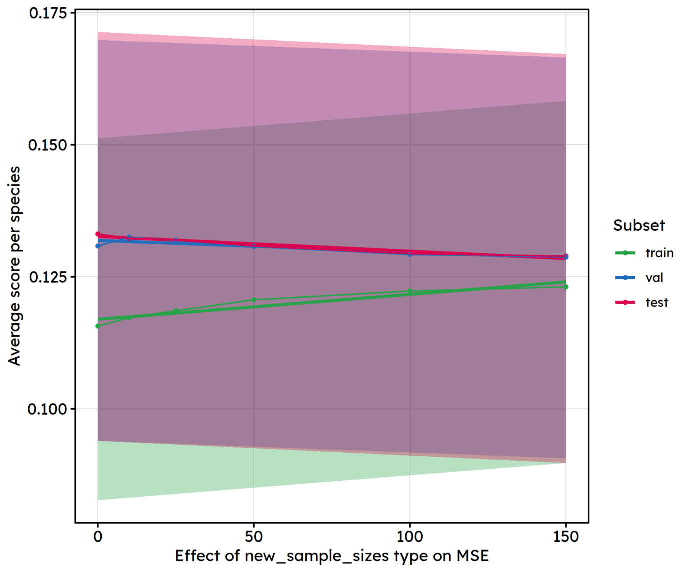
Distribution of per-fold differences (species-averaged within each fold, across k_fold).
Reference: train_sizes=125, r_effects=none, new_sample_sizes=0, strategies=3-part-simplified-uncertainty, hmsc_xformulas=6..
Compared with: train_sizes=125, r_effects=none, new_sample_sizes=see x-axis, strategies=3-part-simplified-uncertainty, hmsc_xformulas=6.

Boxplot improvement



Number of species per box: 15 .
Reference: train_sizes=125, r_effects=none, new_sample_sizes=0, strategies=3-part-simplified-uncertainty, hmsc_xformulas=6..
Compared with: train_sizes=125, r_effects=none, new_sample_sizes=see x-axis, strategies=3-part-simplified-uncertainty, hmsc_xformulas=6.

Estimated effects



Comparison of scores per subset, k_fold mean + 95% CI.
Model type: train_sizes=125, r_effects=none, new_sample_sizes=see x-axis, strategies=3-part-simplified-uncertainty, hmsc_xformulas=.

Fitted model:

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: score ~ loop_element + (1 | k_fold) + (1 | species)
   Data: subset(scores_df, dataset == subset)

REML criterion at convergence: -5856.3

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.1375 -0.6922 -0.0706  0.6391  3.3016

Random effects:
 Groups   Name        Variance  Std.Dev.
 species  (Intercept) 5.824e-03 0.076314
 k_fold   (Intercept) 6.190e-06 0.002488
 Residual             7.211e-05 0.008492
Number of obs: 900, groups:  species, 15; k_fold, 10

Fixed effects:
               Estimate Std. Error         df t value Pr(>|t|)
(Intercept)   1.327e-01  1.972e-02  1.405e+01   6.726 9.52e-06 ***
loop_element -2.791e-05  5.314e-06  8.750e+02  -5.253 1.88e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusions

Overall, these results show that the more we add new samples to the training set, the better the models perform on tests sets and validation sets.

Looking at "Boxplot improvement" plots (for "GF", "SU", and "3PSU" strategies), the effect seems linear when adding new samples using the "SU" strategy, while improvements for "GF" and "3PSU" strategies reach a plateau after 50 new samples added, with no clear progress when adding more samples (100 or 150). This is very interesting as it indicates that, if using an optimal sampling strategy, adding more than 50-100 samples to our original dataset is not very informative.

Overall discussion

Other effects (not shown in this report but with figures available in this folder) were investigated.

Quick summary of results here: - Training size has a positive effect on test performance. This is the strongest effect measured. - Sampling strategy: "GF" and "SU" perform significantly better on test data than other strategies. - Number of new samples (independent of overall training size) and number of explicative variables both show statistically detectable but modest effects in practice. - Explicit spatial random effects significantly impact training scores and validation scores but not test scores.

It should also be noted that, in our models, **species random effect is high**. As our aim is to generalise the predictions beyond the 15 species used here, this is fine for our usage. However it should be kept in mind that these results are highly species-dependant.

K-folds random effects are negligible, which is expected and a good thing, meaning that the partitioning of the dataset has very little effect on scores.

Overall, increasing the number of training samples is the dominant lever for improving HMSC model performance, with returns flattening beyond roughly 50-100 sampled plots (with "BAU" strategy for base model). However, this is only true for an initial random sampling of the STOC dataset, what if the initial sampled agricultural plots were not randomly chosen ? (i.e. in BIODICAPT, some explicative variables could be over-represented, which would skew the distributions). We'll have to change the dataset to see that!

1. Ovaskainen O, Tikhonov G, Norberg A, Blanchet FG, Duan L, Dunson D, et al. How to make more out of community data? A conceptual framework and its implementation as models and software. Ecology Letters [Internet]. 2017;20(5):561–576. Available from: https://doi.org/10.1111/ele.12757

2. Tikhonov G, Opedal ØH, Abrego N, Lehikoinen A, Jonge MMJ de, Oksanen J, et al. Joint species distribution modelling with the r-package Hmsc. Methods in Ecology and Evolution [Internet]. 2020 [cited 2020 Nov 23];11(3):442–447. Available from: https://besjournals.onlinelibrary.wiley.com/doi/abs/10.1111/2041-210X.13345

3. Vallé C, Poggiato G, Thuiller W, Jiguet F, Princé K, Le Viol I. Species associations in joint species distribution models: From missing variables to conditional predictions. Journal of Biogeography [Internet]. 2024 Feb [cited 2026 June 12];51(2):311–324. Available from: https://onlinelibrary.wiley.com/doi/10.1111/jbi.14752

4. Mondain-Monval T, Pocock M, Rolph S, August T, Wright E, Jarvis S. Adaptive sampling by citizen scientists improves species distribution model performance: A simulation study. Methods in Ecology and Evolution [Internet]. 2024 July [cited 2026 May 22];15(7):1206–1220. Available from: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210X.14355